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Forming duct structure with overbraided woven fiber sleeve

Abstract

A method is provided for manufacturing. During this method, a first woven fiber sleeve is disposed over a first mandrel to provide a first overbraided mandrel. The first woven fiber sleeve is wrapped circumferentially around the first mandrel. An opening is formed through a side of the first woven fiber sleeve. A second woven fiber sleeve is disposed over a second mandrel to provide a second overbraided mandrel. The second woven fiber sleeve is wrapped circumferentially around the second mandrel. The second overbraided mandrel is arranged with the first overbraided mandrel. The second woven fiber sleeve engages the side of the first woven fiber sleeve. A second overbraid first end is disposed at and extends circumferentially around the opening. A polymer material is disposed with the first woven fiber sleeve and the second woven fiber sleeve to provide a duct structure.

Inventors:	Davis; Jennifer (Carlsbad, CA)
Applicant:	Rohr, Inc. (Chula Vista, CA)
Family ID:	1000008777924
Assignee:	Rohr, Inc. (Chula Vista, CA)
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Primary Examiner: Daniels; Matthew J

Attorney, Agent or Firm: Getz Balich LLC

Background/Summary

BACKGROUND

1. Technical Field

(1) This disclosure relates generally to manufacturing a fiber-reinforced composite duct structure.

2. Background Information

(2) An aircraft propulsion system may include a duct structure for delivering air to an engine such as a turboprop gas turbine engine. The duct structure may also include a bypass duct. Various types of duct structures and methods for forming those duct structures are known in the art. While these known duct structures and methods for forming duct structures have various benefits, there is still room in the art for improvement.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, a method is provided for manufacturing. During this method, a first woven fiber sleeve is disposed over a first mandrel to provide a first overbraided mandrel. The first woven fiber sleeve is wrapped circumferentially around the first mandrel. The first woven fiber sleeve extends longitudinally along the first mandrel between a first overbraid first end and a first overbraid second end. An opening is formed through a side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end. A second woven fiber sleeve is disposed over a second mandrel to provide a second overbraided mandrel. The second woven fiber sleeve is wrapped circumferentially around the second mandrel. The second woven fiber sleeve extends longitudinally along the second mandrel between a second overbraid first end and a second overbraid second end. The second overbraided mandrel is arranged with the first overbraided mandrel. The second woven fiber sleeve engages the side of the first woven fiber sleeve. The second overbraid first end is disposed at and extends circumferentially around the opening. A polymer material is disposed with the first woven fiber sleeve and the second woven fiber sleeve to provide a duct structure.

(4) According to another aspect of the present disclosure, another method is provided for manufacturing. During this method, a first woven fiber sleeve is disposed over a first mandrel to provide a first overbraided mandrel. The first woven fiber sleeve is wrapped circumferentially around the first mandrel. The first woven fiber sleeve extends longitudinally along the first mandrel between a first overbraid first end and a first overbraid second end. An opening is formed through a side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end. A second woven fiber sleeve is over a second mandrel to provide a second overbraided mandrel. The second woven fiber sleeve is wrapped circumferentially around the second mandrel. The second woven fiber sleeve extends longitudinally along the second mandrel between a second overbraid first end and a second overbraid second end. The second overbraided mandrel is inserted through the opening and into an aperture in the first mandrel. The second woven fiber sleeve engages the side of the first woven fiber sleeve. The the second overbraid first end is disposed at the opening. A polymer material is disposed with the first woven fiber sleeve and the second woven fiber sleeve to provide a duct structure.

(5) According to still another aspect of the present disclosure, another method is provided for manufacturing. During this method, a first woven fiber sleeve is disposed over a first mandrel to provide a first overbraided mandrel. The first woven fiber sleeve is wrapped circumferentially around the first mandrel. The first woven fiber sleeve extends longitudinally along the first mandrel between a first overbraid first end and a first overbraid second end. An opening is formed through a side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end. A thermoset material is disposed with the first woven fiber sleeve to provide a first duct section. A second woven fiber sleeve is disposed over a second mandrel to provide a second overbraided mandrel. The second woven fiber sleeve is wrapped circumferentially around the second mandrel. The second woven fiber sleeve extends longitudinally along the second mandrel between a second overbraid first end and a second overbraid second end. The thermoset material is disposed with the second woven fiber sleeve to provide a second duct section. The second duct section is arranged with the first duct section. The second woven fiber sleeve of the second duct section engages the side of the first woven fiber sleeve of the first duct section. The second overbraid first end is disposed at and extends circumferentially around the opening. A duct structure is formed by attaching the second duct section to the first duct section.

(6) The thermoset material of the second duct section may be co-cured with the thermoset material of the first duct section during the attaching.

(7) The second duct section may be mechanically fastened to the first duct section during the attaching.

(8) The duct structure may include an inlet, a first outlet and a second outlet. The duct structure

may fluidly couple and branch out from the inlet to the first outlet and the second outlet.

(9) The first woven fiber sleeve may form the inlet at the first overbraid first end. The first woven fiber sleeve may form the first outlet at the first overbraid second end. The second woven fiber sleeve may form the second outlet at the second overbraid second end.

(10) The second woven fiber sleeve of the second overbraided mandrel may include a base section and a mount section. The base section may be wrapped circumferentially around the second mandrel. The base section may extend longitudinally along the second mandrel between the second overbraid first end and the second overbraid second end. The mount section may be disposed at the second overbraid first end and may project out from the base section.

(11) The mount section may be abutted against the side of the first woven fiber sleeve during the arranging of the second overbraided mandrel.

(12) The mount section may form an annular flange around the opening.

(13) The method may also include: disposing a third woven fiber sleeve over a third mandrel to provide a third overbraided mandrel, the third woven fiber sleeve wrapped circumferentially around the third mandrel, and the third woven fiber sleeve extending longitudinally along the third mandrel between a third overbraid first end and a third overbraid second end; and arranging the third overbraided mandrel with the first overbraided mandrel, the third woven fiber sleeve engaging the side of the first woven fiber sleeve, and the third overbraid first end disposed at a second opening in the side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end; the disposing of the polymer material further including disposing the polymer material with the third woven fiber sleeve to provide the duct structure.

(14) The arranging of the third overbraided mandrel may include inserting the third overbraided mandrel through the second opening and into an aperture in the first mandrel.

(15) The third woven fiber sleeve of the third overbraided mandrel may include a base section and a mount section. The base section may be wrapped circumferentially around the third mandrel. The base section may extend longitudinally along the third mandrel between the third overbraid first end and the third overbraid second end. The mount section may be disposed at the third overbraid first end and may project out from the base section.

(16) The mount section may be abutted against the side of the first woven fiber sleeve during the arranging of the third overbraided mandrel.

(17) The mount section may form an annular flange around the second opening.

(18) The method may also include: arranging one or more electric heating elements with at least one of the first woven fiber sleeve or the second woven fiber sleeve; the one or more electric heating elements configured for an anti-icing system for the duct structure.

(19) The method may also include: providing an acoustic structure for the duct structure; the acoustic structure including a perforated face skin, a back skin and a cellular core between and connected to the perforated face skin and the back skin; and the perforated face skin including a portion of the first woven fiber sleeve.

(20) The polymer material may be configured from or otherwise include thermoset material.

(21) The disposing of the polymer material may include: infusing the polymer material into the first woven fiber sleeve and the second woven fiber sleeve; and curing the polymer material to provide the duct structure.

(22) The duct structure may be configured for an aircraft propulsion system.

(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side schematic illustration of an aircraft propulsion system.
- (2) FIG. 2 is a side schematic illustration of an engine for the aircraft propulsion system.
- (3) FIG. 3 is a perspective illustration of a duct assembly for the aircraft propulsion system.
- (4) FIG. 4 is a sectional illustration of a duct structure.
- (5) FIG. 5 is a partial illustration of the duct structure at an outlet from an engine duct flowpath.
- (6) FIGS. 6A-C are schematic cross-sectional illustrations of various ducts of the duct structure.
- (7) FIG. 7 is a perspective illustration of a base duct section of the duct structure.
- (8) FIG. 8 is a perspective illustration of a bypass duct section of the duct structure.
- (9) FIG. 9 is a perspective illustration of a shaft duct section of the duct structure.
- (10) FIG. 10 is a flow diagram of a method for manufacturing a duct structure.
- (11) FIG. 11 is a sectional illustration of a base duct section overbraided mandrel.
- (12) FIG. 12 is a perspective illustration of a portion of the woven fiber sleeve.
- (13) FIG. 13 is a sectional illustration of a bypass duct section overbraided mandrel.
- (14) FIG. 14 is a sectional illustration of a shaft duct section overbraided mandrel.
- (15) FIG. 15 is a sectional illustration of the overbraided mandrels with tooling.
- (16) FIG. 16 is a perspective illustration of the duct assembly with one or more acoustic structures.
- (17) FIG. 17 is a partial cutaway illustration of one of the acoustic structures.

DETAILED DESCRIPTION

(18) The present disclosure includes methods for manufacturing a duct structure from fiber-reinforced composite material. This duct structure may be configured for an aircraft such as an airplane or another manned or unmanned aerial vehicle. The duct structure, for example, may be configured for a propulsion system of the aircraft. The present disclosure, however, is not limited to such an exemplary aircraft application. The duct structure, for example, may alternatively be configured for use in a part or system of the aircraft outside of (or in combination with) the aircraft propulsion system. Furthermore, the duct structure is not limited to aircraft applications in general. The duct structure, for example, may be configured for any application which would benefit from use of a (e.g., monolithic) fiber-reinforced composite duct structure manufactured as described below. However, for ease of description, the duct structure may be described with respect to an aircraft propulsion system.

(19) FIG. 1 illustrates an exemplary aircraft propulsion system 20. This aircraft propulsion system 20 includes an engine 22 and a duct assembly 24. The aircraft propulsion system 20 also includes a nacelle 26 configured to at least partially (or completely) house and provide an aerodynamic cover for the engine 22 and/or the duct assembly 24.

(20) Referring to FIG. 2, the engine 22 may be configured as an open rotor gas turbine engine such as a turboprop gas turbine engine. The engine 22 of FIG. 2, for example, includes a propeller section 28, a compressor section 29, a combustor section 30 and a turbine section 31. The turbine section 31 may include a high pressure turbine (HPT) section 31A and a low pressure turbine (LPT) section 31B; e.g., a power turbine section. A core 34 of the engine 22 includes the compressor section 29, the combustor section 30, the HPT section 31A and the LPT section 31B.

(21) The engine sections 29-31B are arranged sequentially along a core flowpath 36 within the engine core 34. This core flowpath 36 extends within the engine core 34 from an upstream airflow inlet 38 into the engine core 34 to a downstream combustion products exhaust 40 from the engine core 34. Here, the airflow inlet 38 is also an airflow inlet into the engine 22 and the exhaust 40 is also a combustion products exhaust from the engine 22; however, the present disclosure is not limited to such an exemplary arrangement.

(22) Each of the engine sections 28, 29, 31A and 31B includes a respective bladed rotor 42-45. Each of these bladed rotors 42-45 includes a plurality of rotor blades arranged circumferentially around and connected to one or more respective rotor disks. The rotor blades, for example, may be

formed integral with or mechanically fastened, welded, brazed, adhered and/or otherwise attached to the respective rotor disk(s). The propeller rotor **42** is connected to a geartrain **48** through a propeller shaft **50**. The geartrain **48** is connected to and driven by the LPT rotor **45** through a low speed shaft **52**. The compressor rotor **43** is connected to and driven by the HPT rotor **44** through a high speed shaft **54**.

(23) During engine operation, air enters the engine core **34** through the airflow inlet **38** and is directed into the core flowpath **36**. The air within the core flowpath **36** may be referred to as “core air”. This core air is compressed by the compressor rotor **43** and directed into a combustion chamber **56** of a combustor in the combustor section **30**. Fuel is injected into the combustion chamber **56** through one or more fuel injectors and mixed with the compressor core air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially cause the HPT rotor **44** and the LPT rotor **45** to rotate. The rotation of the HPT rotor **44** drives rotation of the compressor rotor **43** and, thus, compression of the air received from the airflow inlet **38**. The rotation of the LPT rotor **45** drives rotation of the propeller rotor **42**. The rotation of the propeller rotor **42** generates forward aircraft thrust by propelling additional air (e.g., outside of the engine core **34** and the nacelle **26** of FIG. **1**) in an aft direction.

(24) The engine **22** is described above as the turboprop gas turbine engine for ease of description. The present disclosure, however, is not limited to such an exemplary engine. The engine **22**, for example, may alternatively be configured as another type of open rotor gas turbine engine, or alternatively a ducted gas turbine engine such as a turbofan or turbojet gas turbine engine. Furthermore, it is also contemplated the engine core **34** may be replaced with another engine powerplant such as, but not limited to, a reciprocating piston engine, a rotary engine or another type of internal combustion engine.

(25) Referring to FIG. **1**, the duct assembly **24** includes an inlet nose lip **58** of the nacelle **26** and the fiber-reinforced composite duct structure **60**. The nose lip **58** is configured to form an airflow inlet **62** into the aircraft propulsion system **20** and its duct assembly **24**. The nose lip **58** of FIG. **1** is arranged at (e.g., on, adjacent or proximate) a forward end of the nacelle **26**. This nose lip **58** may be vertically below (e.g., relative to gravity) and aft/downstream of the propeller rotor hub. The nose lip **58** of the present disclosure, however, is not limited to such an exemplary relative position to the propeller rotor **42**.

(26) Referring to FIGS. **3** and **4**, the duct structure **60** includes a plurality of ducts such as, but not limited to, an inlet duct **64**, an engine duct **65**, a bypass duct **66** and a shaft duct **67**. This duct structure **60** may be configured as a monolithic body. The duct structure members (e.g., the ducts **64-67**), for example, may be molded and/or otherwise formed integral together to configure the duct structure **60** as a single, unitary body. By contrast, a non-monolithic body may include a plurality of bodies which are discretely formed and subsequently (e.g., mechanically) fastened or welded to one another. By forming the duct structure **60** as the monolithic body, an overall weight and/or complexity of the duct structure **60** may be reduced. The present disclosure, however, is not limited to such an exemplary monolithic arrangement as described below in further detail.

(27) The duct structure **60** may be constructed from or otherwise include a polymer material. This polymer material is structurally reinforced with fiber reinforcement. The fiber reinforcement, for example, may be embedded within a matrix of the polymer material. The polymer material may be a thermoset such as, but not limited to, epoxy or toughened epoxy. However, it is contemplated the polymer material may alternatively be a thermoplastic such as, but not limited to, polyether ether ketone (PEEK), polyaryletherketone (PAEK), polyether ketone ketone (PEKK) or polyphenylene sulfide (PPS). However, for ease of description, the polymer material may be described below as the thermoset. The fiber reinforcement may be or otherwise include fiberglass fibers, carbon fibers, aramid (e.g., Kevlar®) fibers, or some combination therewith. The present disclosure, however, is not limited to the foregoing exemplary duct structure materials.

(28) The inlet duct **64** of FIG. **4** has an internal flowpath **70**; e.g., a bore. The inlet duct **64** and its

flowpath **70** extend longitudinally along a longitudinal centerline **72** of a base duct section **74** of the duct structure **60** and its duct structure members (e.g., the ducts **64** and **65**) from a forward, upstream end **76** of the duct structure **60** to a transition **78** to (e.g., an interface with) the engine duct **65** and the bypass duct **66**. At the structure upstream end **76** of FIG. **1**, the inlet duct **64** is attached (e.g., mechanically fastened or bonded) to the nose lip **58**. The inlet duct flowpath **70** is also fluidly coupled with the airflow inlet **62**. At the inter-duct transition **78** of FIG. **4**, the inlet duct **64** is connected to (e.g., formed integral with or otherwise attached to) the engine duct **65** and the bypass duct **66**. The inlet duct flowpath **70** is also fluidly coupled with (e.g., in parallel) an inlet **80** of and into the engine duct **65** and an inlet **82** of and into the bypass duct **66**. With this arrangement, the engine duct **65** and the bypass duct **66** branch off from and are fluidly fed by the inlet duct **64**.

(29) The engine duct **65** has an internal flowpath **84**; e.g., a bore. The engine duct **65** and its flowpath **84** extend longitudinally along the longitudinal centerline **72** from the engine duct inlet **80** to an outlet **86** of and from the engine duct **65**. This engine duct outlet **86** is disposed at an aft, downstream end **88** of the engine duct **65**. The engine duct outlet **86** is configured to fluidly couple the engine duct **65** and its flowpath **84** to the engine **22** and its airflow inlet **38**; see FIG. **1**.

Referring to FIG. **5**, the engine duct outlet **86** may be an annular orifice. The engine duct outlet **86** of FIG. **5**, for example, extends radially between and to an exterior surface of the shaft duct **67** and an interior surface of the engine duct **65**. The engine duct outlet **86** of FIG. **5** also extends within the duct structure **60** and its engine duct **65** circumferentially about (e.g., completely around) the longitudinal centerline **72**. The engine duct outlet **86** may thereby circumscribe the shaft duct **67**, and the engine duct **65** may circumscribe the engine duct outlet **86**. In other embodiments, however, it is contemplated the engine duct outlet **86** may be formed by and radially between the engine duct **65** and another duct where, for example, the shaft duct **67** is recessed from the engine duct downstream end **88** and the other duct provides an extension for the shaft duct **67**.

(30) Referring to FIG. **4**, after branching off from the inlet duct **64**, the engine duct **65** may turn away from the bypass duct **66**; e.g., vertically upwards and then horizontal in FIG. **4**. The engine duct **65** and its longitudinal centerline **72** of FIG. **4**, for example, follow a non-straight (e.g., a curved, a wavy, a splined, etc.) trajectory from the inlet duct **64** and to the engine duct outlet **86**. With this arrangement, a forward, upstream portion of the engine duct **65** and its flowpath **84** may be angularly offset from and downstream of the inlet duct **64** and its flowpath **70**. The present disclosure, however, is not limited to such an exemplary geometry.

(31) The bypass duct **66** has an internal flowpath **90**; e.g., a bore. The bypass duct **66** and its flowpath **90** extend longitudinally along a longitudinal centerline **92** of a bypass duct section **94** of the duct structure **60** and its bypass duct **66** from the bypass duct inlet **82** to an outlet **96** of and from the bypass duct **66**. This bypass duct outlet **96** is disposed at an aft, downstream end **98** of the bypass duct **66**. Referring to FIG. **1**, the bypass duct outlet **96** may be configured as a bypass exhaust **100** from the duct structure **60** into an environment **102** external to the aircraft propulsion system **20** and its nacelle **26**; e.g., an external environment, an ambient environment, etc.

(32) Referring to FIG. **4**, after branching off from the inlet duct **64**, the bypass duct **66** may turn away from the engine duct **65**; e.g., horizontally and then vertically downwards in FIG. **4**. The bypass duct **66** and its longitudinal centerline **92** of FIG. **4**, for example, follow a non-straight (e.g., a curved, a wavy, a splined, etc.) trajectory away from the inlet duct **64** and to the bypass duct outlet **96**/the bypass exhaust **100**. With this arrangement, a forward, upstream portion of the bypass duct **66** and its flowpath **90** may be substantially aligned in a straight line with and downstream of the inlet duct **64** and its flowpath **70**. The present disclosure, however, is not limited to such an exemplary geometry.

(33) At the inter-duct transition **78**, referring to FIG. **6A-C**, each of the duct flowpaths **70**, **84**, **90** has a cross-sectional area (schematically shown). The inlet duct cross-sectional area may be sized greater than the engine duct cross-sectional area and the bypass duct cross-sectional area. The inlet

duct cross-sectional area, for example, may be exactly equal to or approximately equal to (e.g., within $\pm 2\%$ of) a sum of the engine duct cross-sectional area and the bypass duct cross-sectional area. The engine duct cross-sectional area, however, may be sized greater than the bypass duct cross-sectional area. The engine duct cross-sectional area, for example, may be between one and one-half times ($1.5\times$) and five or ten times ($5\times$, $10\times$) the bypass duct cross-sectional area. The inlet duct **64** of FIG. **4** may thereby be configured to direct a majority of fluid received from the nose lip **58** and its airflow inlet **62** (see FIG. **1**) into the engine duct **65** and its flowpath **84** over the bypass duct **66**. The present disclosure, however, is not limited to the foregoing exemplary dimensional relationships.

(34) The shaft duct **67** has an internal bore **104**. The shaft duct **67** and its internal bore **104** extend longitudinally along a longitudinal centerline **106** (e.g., a centerline axis) a shaft duct section **108** of the duct structure **60** and its shaft duct **67** from a forward, upstream end **110** of the shaft duct **67** to an aft, downstream end **112** of the shaft duct **67**. The shaft duct upstream end **110** is disposed at a sidewall **114** of the engine duct **65**. At this shaft duct upstream end **110**, a sidewall **116** of the shaft duct **67** is connected to the engine duct sidewall **114** and extends circumferentially about (e.g., completely around) a shaft opening **118** through the engine duct sidewall **114**. The shaft duct downstream end **112** is disposed at the engine duct downstream end **88**. With this arrangement, the internal bore **104** forms an open passageway through the shaft duct **67** as well as through the duct structure **60**. This open passageway/the internal bore **104** is configured to receive a shaft for the aircraft propulsion system **20**. The low speed shaft **52** of FIG. **1**, for example, may extend axially through the open passageway/the internal bore of the shaft duct **67**. In other embodiments, however, it is contemplated the propeller shaft **50** (or another shaft) may alternatively extend axially through the open passageway/the internal bore of the shaft duct **67**. With this arrangement, the duct structure **60** may direct air into an axial end of the engine core **34** without requiring offsetting of the propeller shaft **50**, etc.

(35) With the foregoing configuration, the duct assembly **24** is configured to receive (e.g., ingest) ambient air through the nose lip **58** from external environment **102**. The duct assembly **24** and its duct structure **60** are configured to direct some of the ambient air to the engine **22** and its airflow inlet **38** through the engine duct **65** to supply (e.g., all or at least some) of the core air. The duct assembly **24** is also configured to bypass some of the ambient air from the engine **22** and its core flowpath **36** into the bypass duct **66**. More particularly, the duct assembly **24** may be configured such that any, substantially all or at least some debris (e.g., foreign object debris (FOD)) which enters the duct assembly **24** with the ambient air through the nose lip **58** is carried with the bypass air and exhausted from the aircraft propulsion system **20** through the bypass exhaust **100** without entering the engine **22** and its core flowpath **36**. It is contemplated, of course, the bypass air flowing through the bypass duct **66** may also or alternatively be used for various other purposes.

(36) The duct structure **60** of FIGS. **3** and **4** may be divided into a plurality of duct sections such as, but not limited to, the base duct section **74**, the bypass duct section **94** and the shaft duct section **108**. FIG. **7** illustrates the base duct section **74** without the other duct sections **94** and **108** for ease of illustration. FIG. **8** illustrates the bypass duct section **94** without the other duct sections **74** and **108** for ease of illustration. FIG. **9** illustrates the shaft duct section **108** without the other duct sections **74** and **94** for ease of illustration.

(37) The base duct section **74** of FIG. **7** is configured as a tubular member (or body) of the duct structure **60**. This base duct section **74** forms the inlet duct **64** and the engine duct **65**. The base duct section **74** of FIG. **7**, for example, extends longitudinally along the longitudinal centerline **72** from the structure upstream end **76** to the engine duct downstream end **88**. The base duct section **74** includes a bypass opening **120** and the shaft opening **118**. The bypass opening **120** is a port (e.g., an orifice) through a sidewall **122** of the base duct section **74**; e.g., a sidewall **124** of the inlet duct **64**. This bypass opening **120** may form the bypass duct inlet **82** (see FIG. **4**). The shaft opening **118** is a port (e.g., an orifice) through the sidewall **122** of the base duct section **74**; e.g., the sidewall **114** of

the engine duct **65**.

(38) The bypass duct section **94** of FIG. **8** is configured as a tubular member (or body) of duct structure **60**. This bypass duct section **94** forms the bypass duct **66**. The bypass duct section **94** of FIG. **8**, for example, extends longitudinally along the longitudinal centerline **92** from a forward, upstream end **126** to the bypass duct downstream end **98**. The bypass duct section **94** of FIG. **8**, in particular, includes a base section **128** (e.g., a tubular sidewall) and a mount section **130** (e.g., an annular flange). The base section **128** forms the bypass duct **66**. The mount section **130** is disposed at the bypass duct upstream end **126**. This mount section **130** projects out (e.g., radially outward) from the base section **128**. The mount section **130** may extend circumferentially about the base section **128**, thereby providing the mount section **130** with a full-hoop annular geometry. Referring to FIGS. **3** and **4**, the mount section **130** is configured to engage (e.g., contact) and connect the bypass duct **66** to the base duct section **74** and its inlet duct **64**. Here, the mount section **130** extends circumferentially about the bypass opening **120**/the bypass duct inlet **82**.

(39) The shaft duct section **108** of FIG. **9** is configured as a tubular member (or body) of the duct structure **60**. This shaft duct section **108** forms the shaft duct **67**. The shaft duct section **108** of FIG. **9**, for example, extends longitudinally along the longitudinal centerline **106** from the shaft duct upstream end **110** to the shaft duct downstream end **112**. The shaft duct section **108** of FIG. **9**, in particular, includes a base section **132** and a mount section **134**; e.g., annular flange. The base section **132** forms the shaft duct **67**. The mount section **134** is disposed at the shaft duct upstream end **110**. This mount section **134** projects out (e.g., radially outward) from the base section **132**. The mount section **134** may extend circumferentially about the base section **132**, thereby providing the mount section **134** with a full-hoop annular geometry. Referring to FIGS. **3** and **4**, the mount section **134** is configured to engage (e.g., contact) and connect the shaft duct **67** to the base duct section **74** and its engine duct **65**. Here, the mount section **134** extends circumferentially around the shaft opening **118**.

(40) While the duct sections **74**, **94** and **108** are shown in FIGS. **7-9** as separate bodies, two or all of these duct sections **74**, **94** and **108** may be formed together as the monolithic body. In such embodiments, the mount sections **130**, **134** may provide reinforced interfaces between the ducts **64** and **66**, **65** and **67**. Of course, in other embodiments, it is contemplated that one or more of these duct sections **74**, **94** and **108** may each be formed as a discrete body and then attached (e.g., mechanically fastened or bonded to) the other respective duct section(s) **74**, **94** and **108**.

(41) FIG. **10** is a flow diagram of a method **1000** for manufacturing a duct structure. For ease of description, the manufacturing method **1000** of FIG. **10** is described with reference to the duct structure **60** of FIGS. **1-9**. The manufacturing method **1000** of the present disclosure, however, is not limited to manufacturing such an exemplary duct structure.

(42) In step **1002**, referring to FIG. **11**, a base duct section woven fiber sleeve **136** is disposed over a base duct section mandrel **138** to provide a base duct section overbraided mandrel **140**. Referring to FIG. **12**, the base duct section woven fiber sleeve **136** may be configured as a circumferentially continuous tubular sleeve formed (e.g., woven) from the fiber reinforcement. The base duct section woven fiber sleeve **136** of FIG. **12**, for example, is configured as a braided sleeve of the fiber reinforcement; e.g., carbon fiber braided sleeve, fiberglass braided sleeve, aramid fiber braided sleeve or the like. Various patterns may be used for weaving (e.g., braiding) the fiber reinforcement to form the base duct section woven fiber sleeve **136**, and the present disclosure is not limited to any particular ones thereof.

(43) Referring to FIG. **11**, during the disposing step **1002**, the base duct section woven fiber sleeve **136** may be slipped longitudinally onto an end of the base duct section mandrel **138**. The base duct section mandrel **138** may thereby project longitudinally into an open end of the base duct section woven fiber sleeve **136**. The base duct section woven fiber sleeve **136** may then be pulled along and (e.g., completely) onto the base duct section mandrel **138** such that, for example, an entire (or substantial) longitudinal length of the base duct section mandrel **138** is covered and overlapped by

the base duct section woven fiber sleeve **136**. In addition, one or more openings **142** and **144** may be formed through a side of the base duct section woven fiber sleeve **136**. These openings **142** and **144** may correspond to the bypass opening **120** and the shaft opening **118**. One or more of these openings **142** and **144** may be formed by manipulating (e.g., pulling apart) the fiber reinforcement of the base duct section woven fiber sleeve **136** and/or cutting the base duct section woven fiber sleeve **136**.

(44) In step **1004**, referring to FIG. **13**, a bypass duct section woven fiber sleeve **146** is disposed over a bypass duct section mandrel **148** to provide a bypass duct section overbraided mandrel **150**. Referring to FIG. **12**, the bypass duct section woven fiber sleeve **146** may be configured as a circumferentially continuous tubular sleeve formed (e.g., woven) from the fiber reinforcement. The bypass duct section woven fiber sleeve **146** of FIG. **12**, for example, is configured as a braided sleeve of the fiber reinforcement; e.g., carbon fiber braided sleeve, fiberglass braided sleeve, aramid fiber braided sleeve or the like. Various patterns may be used for weaving (e.g., braiding) the fiber reinforcement to form the bypass duct section woven fiber sleeve **146**, and the present disclosure is not limited to any particular ones thereof.

(45) Referring to FIG. **13**, during the disposing step **1004**, the bypass duct section woven fiber sleeve **146** may be slipped longitudinally onto an end of the bypass duct section mandrel **148**. The bypass duct section mandrel **148** may thereby project longitudinally into an open end of the bypass duct section woven fiber sleeve **146**. The bypass duct section woven fiber sleeve **146** may then be pulled along and (e.g., completely) onto the bypass duct section mandrel **148** such that, for example, an entire (or substantial) longitudinal length of the bypass duct section mandrel **148** is covered and overlapped by the bypass duct section woven fiber sleeve **146**.

(46) At least one end portion of the bypass duct section woven fiber sleeve **146** is manipulated to provide the bypass duct section woven fiber sleeve **146** with an end mount section **152**. This end portion of the bypass duct section woven fiber sleeve **146**, for example, may be flared outward from a base section **154** of the bypass duct section woven fiber sleeve **146**. More particularly, the fiber reinforcement at the end portion of the bypass duct section woven fiber sleeve **146** may be splayed or otherwise spread apart (e.g., without requiring cutting of the bypass duct section woven fiber sleeve **146**) to respectively form the end mount section **152**. This end mount section **152** projects out from and circumscribes the base section **154**. Here, the base section **154** is wrapped circumferentially around the bypass duct section mandrel **148** and extends longitudinally along the bypass duct section mandrel **148** between and to opposing longitudinal overbraid ends **151** and **153** of, for example, the bypass duct section overbraided material.

(47) In step **1006**, referring to FIG. **14**, a shaft duct section woven fiber sleeve **156** is disposed over a shaft duct section mandrel **158** to provide a shaft duct section overbraided mandrel **160**. Referring to FIG. **12**, the shaft duct section woven fiber sleeve **156** may be configured as a circumferentially continuous tubular sleeve formed (e.g., woven) from the fiber reinforcement. The shaft duct section woven fiber sleeve **156** of FIG. **12**, for example, is configured as a braided sleeve of the fiber reinforcement; e.g., carbon fiber braided sleeve, fiberglass braided sleeve, aramid fiber braided sleeve or the like. Various patterns may be used for weaving (e.g., braiding) the fiber reinforcement to form the shaft duct section woven fiber sleeve **156**, and the present disclosure is not limited to any particular ones thereof.

(48) Referring to FIG. **14**, during the disposing step **1006**, the shaft duct section woven fiber sleeve **156** may be slipped longitudinally onto an end of the shaft duct section mandrel **158**. The shaft duct section mandrel **158** may thereby project longitudinally into an open end of the shaft duct section woven fiber sleeve **156**. The shaft duct section woven fiber sleeve **156** may then be pulled along and (e.g., completely) onto the shaft duct section mandrel **158** such that, for example, an entire (or substantial) longitudinal length of the shaft duct section mandrel **158** is covered and overlapped by the shaft duct section woven fiber sleeve **156**.

(49) At least one end portion of the shaft duct section woven fiber sleeve **156** is manipulated to

provide the shaft duct section woven fiber sleeve **156** with an end mount section **162**. This end portion of the shaft duct section woven fiber sleeve **156**, for example, may be flared outward from a base section **164** of the shaft duct section woven fiber sleeve **156**. More particularly, the fiber reinforcement at the end portion of the shaft duct section woven fiber sleeve **156** may be splayed or otherwise spread apart (e.g., without requiring cutting of the shaft duct section woven fiber sleeve **156**) to respectively form the end mount section **162**. This end mount section **162** projects out from and circumscribes the base section **164**. Here, the base section **164** is wrapped circumferentially around the shaft duct section mandrel **158** and extends longitudinally along the shaft duct section mandrel **158** between and to opposing longitudinal overbraid ends **166** and **168** of, for example, the shaft duct section overbraided material.

(50) In step **1008**, referring to FIG. **15**, the overbraided mandrels **140**, **150** and **160** are arranged together. The bypass duct section overbraided mandrel **150**, for example, is disposed next to a side of (and outside of) the base duct section overbraided mandrel **140**. Here, the bypass duct section woven fiber sleeve **146** engages (e.g., contacts) the base duct section woven fiber sleeve **136**. The overbraid first end **156** and the end mount section **152** are disposed at and extend circumferentially about the bypass opening **120**.

(51) The shaft duct section overbraided mandrel **160** is inserted through the opening **144** and into an aperture **170** (see also FIG. **11**) in the base duct section mandrel **138**. Here, the base section **164** of the shaft duct section woven fiber sleeve **156** is disposed substantially within the aperture **170**. The shaft duct section woven fiber sleeve **156** engages (e.g., contacts) the base duct section woven fiber sleeve **136**. The overbraid first end **166** and the end mount section **162** are disposed at the opening **144**; e.g., outside of the base duct section overbraided mandrel **140**.

(52) In step **1010**, the arrangement of overbraided mandrels **140**, **150** and **160** are arranged with tooling **172**. This tooling **172** may be configured as or otherwise include an exterior mold **174**. Here, the overbraided mandrels **140**, **150** and **160** may be disposed within an internal cavity of the exterior mold **174**. The exterior mold **174** may thereby substantially or completely overlap and circumscribe an exterior of the overbraided material. To facilitate arrangement of the overbraided mandrels **140**, **150** and **160** into the internal cavity, the exterior mold **174** may have a segmented body. The exterior mold **174**, for example, may be formed from two separable halves.

(53) In step **1012**, the polymer material is disposed with the woven fiber sleeves **136**, **146** and **156** to provide the duct structure **60**. The polymer material, for example, may be infused into the fiber reinforcement of the woven fiber sleeves **136**, **146** and **156**, for example, following (or during or before) the step **1010**. Liquid polymer material, for example, may be injected into the exterior mold **174** until the fiber reinforcement of the woven fiber sleeves **136**, **146** and **156** are embedded within a matrix of the polymer material. Where the polymer material is the thermoset, the polymer material may be cured under an elevated temperature and/or pressure between the molding members **138**, **148**, **158** and **174**. Where the polymer is the thermoplastic, the polymer material may be set (e.g., solidified) under pressure between the molding members **138**, **148**, **158** and **174**. Following this formation of the duct structure **60**, the molding members **138**, **148**, **158** and **174** are removed to liberate the formed duct structure **60**.

(54) In some embodiments, any one or more or all of the mandrels **138**, **148** and **158** may each be formed from a relatively compliant material to facilitate removal. Each mandrel **138**, **148**, **158**, for example, may be constructed from or otherwise include thermoplastic or low melt metallic material. Each mandrel **138**, **148**, **158** may alternatively be constructed from a washout sand, where the washout sand may be removed by a dissolvable material. Of course, various other techniques may also or alternatively be employed to facilitate removal of the mandrels **138**, **148** and **158** from an interior of the duct structure **60**.

(55) In some embodiments, referring to FIG. **4**, one or more or all of the duct sections **74**, **94** and **108** may each be formed from a single woven fiber sleeve. In other embodiments, one or more or all of the duct sections **74**, **94** and **108** may also or alternatively each be formed from multiple

woven fiber sleeves; e.g., where one woven fiber sleeve is disposed over another woven fiber sleeve.

(56) In some embodiments, referring to FIG. 3, the duct structure **60** may include one or more electric heating elements **176**; e.g., carbon nano-tube heaters. These electric heating elements **176** may be woven into a respective woven fiber sleeve, or placed between an overlapping set (e.g., stack) of the woven fiber sleeves. These electric heating elements **176** may be arranged at various locations along the duct structure **60**; e.g., at and/or about the nose lip **58**, at and/or about a splitter **178** between the engine duct **65** and the bypass duct **66** (see also FIG. 4).

(57) In some embodiments, referring to FIG. 16, the duct structure **60** may also include one or more acoustic structures **180**; e.g., acoustic panels. Referring to FIG. 17, each acoustic structure **180** includes a perforated face skin **182**, a (e.g., non-perforated) back skin **184** and a cellular core **186**; e.g., a honeycomb or other porous core. The face skin **182** may be formed from a perforated portion of the sidewall **114**, **122**, **124**, **128** of the duct structure **60**; e.g., a perforated portion of the sidewall **124** of the inlet duct **64**. The cellular core **186** is arranged (e.g., radially) between and is connected (e.g., bonded) to the face skin **182** and the back skin **184**. This acoustic structure **180** may be formed onto the inlet duct **64** (and/or another portion of the duct structure **60**) after the respective duct is formed. Alternatively, the back skin **184** and the cellular core **186** may be laid up with the woven fiber sleeve and co-cured to form the acoustic structure **180** with the respective duct in the monolithic body. The present disclosure, however, is not limited to any particular acoustic structure formation techniques. The present disclosure is also not limited to any particular cellular core type or configuration.

(58) While various embodiments of the present invention have been disclosed, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the invention. For example, the present invention as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present invention that some or all of these features may be combined with any one of the aspects and remain within the scope of the invention. Accordingly, the present invention is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A method for manufacturing, comprising: disposing a first woven fiber sleeve over a first mandrel to provide a first overbraided mandrel, the first woven fiber sleeve wrapped circumferentially around the first mandrel, the first woven fiber sleeve extending longitudinally along the first mandrel between a first overbraid first end and a first overbraid second end, and an opening formed through a side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end; disposing a second woven fiber sleeve over a second mandrel to provide a second overbraided mandrel, the second woven fiber sleeve wrapped circumferentially around the second mandrel, and the second woven fiber sleeve extending longitudinally along the second mandrel between a second overbraid first end and a second overbraid second end; arranging the second overbraided mandrel with the first overbraided mandrel, the second woven fiber sleeve engaging the side of the first woven fiber sleeve, and the second overbraid first end disposed at and extending circumferentially around the opening; disposing a polymer material with the first woven fiber sleeve and the second woven fiber sleeve to provide a duct structure; and providing an acoustic structure for the duct structure, the acoustic structure including a perforated face skin, a back skin and a cellular core between and connected to the perforated face skin and the back skin, and the perforated face skin including a portion of the first woven fiber sleeve.

2. The method of claim 1, wherein the duct structure includes an inlet, a first outlet and a second

outlet; and the duct structure fluidly couples and branches out from the inlet to the first outlet and the second outlet.

3. The method of claim 2, wherein the first woven fiber sleeve forms the inlet at the first overbraid first end; the first woven fiber sleeve forms the first outlet at the first overbraid second end; and the second woven fiber sleeve forms the second outlet at the second overbraid second end.

4. The method of claim 1, wherein the second woven fiber sleeve of the second overbraided mandrel includes a base section and a mount section; the base section is wrapped circumferentially around the second mandrel, and the base section extends longitudinally along the second mandrel between the second overbraid first end and the second overbraid second end; and the mount section is disposed at the second overbraid first end and projects out from the base section.

5. The method of claim 4, wherein the mount section is abutted against the side of the first woven fiber sleeve during the arranging of the second overbraided mandrel.

6. The method of claim 4, wherein the mount section forms an annular flange around the opening.

7. The method of claim 1, further comprising: disposing a third woven fiber sleeve over a third mandrel to provide a third overbraided mandrel, the third woven fiber sleeve wrapped circumferentially around the third mandrel, and the third woven fiber sleeve extending longitudinally along the third mandrel between a third overbraid first end and a third overbraid second end; and arranging the third overbraided mandrel with the first overbraided mandrel, the third woven fiber sleeve engaging the side of the first woven fiber sleeve, and the third overbraid first end disposed at a second opening in the side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end; the disposing of the polymer material further comprising disposing the polymer material with the third woven fiber sleeve to provide the duct structure.

8. The method of claim 7, wherein the third woven fiber sleeve of the third overbraided mandrel includes a base section and a mount section; the base section is wrapped circumferentially around the third mandrel, and the base section extends longitudinally along the third mandrel between the third overbraid first end and the third overbraid second end; the mount section is disposed at the third overbraid first end and projects out from the base section.

9. The method of claim 8, wherein the mount section is abutted against the side of the first woven fiber sleeve during the arranging of the third overbraided mandrel.

10. The method of claim 8, wherein the mount section forms an annular flange around the second opening.

11. The method of claim 1, further comprising: arranging one or more electric heating elements with at least one of the first woven fiber sleeve or the second woven fiber sleeve; the one or more electric heating elements configured for an anti-icing system for the duct structure.

12. The method of claim 1, wherein the polymer material comprises thermoset material.

13. The method of claim 1, wherein the disposing of the polymer material comprises infusing the polymer material into the first woven fiber sleeve and the second woven fiber sleeve; and curing the polymer material to provide the duct structure.

14. The method of claim 1, wherein the duct structure is configured for an aircraft propulsion system.

15. A method for manufacturing, comprising: disposing a first woven fiber sleeve over a first mandrel to provide a first overbraided mandrel, the first woven fiber sleeve wrapped circumferentially around the first mandrel, the first woven fiber sleeve extending longitudinally along the first mandrel between a first overbraid first end and a first overbraid second end, and an opening formed through a side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end; disposing a second woven fiber sleeve over a second mandrel to provide a second overbraided mandrel, the second woven fiber sleeve wrapped circumferentially around the second mandrel, and the second woven fiber sleeve extending longitudinally along the second mandrel between a second overbraid first end and a second

overbraid second end; arranging the second overbraided mandrel with the first overbraided mandrel, the second woven fiber sleeve engaging the side of the first woven fiber sleeve, and the second overbraid first end disposed at and extending circumferentially around the opening; disposing a third woven fiber sleeve over a third mandrel to provide a third overbraided mandrel, the third woven fiber sleeve wrapped circumferentially around the third mandrel, and the third woven fiber sleeve extending longitudinally along the third mandrel between a third overbraid first end and a third overbraid second end; arranging the third overbraided mandrel with the first overbraided mandrel, the third woven fiber sleeve engaging the side of the first woven fiber sleeve, and the third overbraid first end disposed at a second opening in the side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end; and disposing a polymer material with the first woven fiber sleeve and the second woven fiber sleeve to provide a duct structure, and the disposing of the polymer material further comprising disposing the polymer material with the third woven fiber sleeve to provide the duct structure; wherein the arranging of the third overbraided mandrel comprises inserting the third overbraided mandrel through the second opening and into an aperture in the first mandrel.

16. A method for manufacturing, comprising: disposing a first woven fiber sleeve over a first mandrel to provide a first overbraided mandrel, the first woven fiber sleeve wrapped circumferentially around the first mandrel, the first woven fiber sleeve extending longitudinally along the first mandrel between a first overbraid first end and a first overbraid second end, and an opening formed through a side of the first woven fiber sleeve longitudinally between the first overbraid first end and the first overbraid second end; disposing a second woven fiber sleeve over a second mandrel to provide a second overbraided mandrel, the second woven fiber sleeve wrapped circumferentially around the second mandrel, and the second woven fiber sleeve extending longitudinally along the second mandrel between a second overbraid first end and a second overbraid second end; inserting the second overbraided mandrel through the opening and into an aperture in the first mandrel, the second woven fiber sleeve engaging the side of the first woven fiber sleeve, and the second overbraid first end disposed at the opening; and disposing a polymer material with the first woven fiber sleeve and the second woven fiber sleeve to provide a duct structure.
